

Conditional Statement

(i) if Statement

```
In [1]: if True:  
        print('Data Science')
```

Data Science

```
In [2]: if False:  
        print('Data Science')  
        print('Bye for now')
```

Bye for now

```
In [3]: if True:  
        print('Data Science')  
        print('Bye for now')
```

Data Science

Bye for now

```
In [4]: x = 4  
        r = x % 2  
  
        if r == 0:  
            print('Even number')
```

Even number

```
In [5]: x = 4  
        r = x % 2  
  
        if r == 0:  
            print('Even number')
```

Even number

```
In [34]: x = 5  
         r = x % 2  
         if r == 0:  
             print('Even number')
```

```
In [8]: x = 5  
        r = x % 2  
  
        if r != 0:  
            print('Odd number')
```

Odd number

```
In [10]: x = 7  
         r = x % 2  
  
         if r == 1:  
             print('Odd number')
```

Odd number

```
In [11]: x = 5
r = x % 2

if r == 0:
    print('Even number')
print('Odd number')
```

Odd number

```
In [12]: x = 4
r = x % 2

if r == 0:
    print('Even number')
print('Odd number')
```

Even number

Odd number

```
In [13]: x = 9
r = x % 2

if r == 0:
    print('Even number')

if r == 1:
    print('Odd number')
```

Odd number

(ii) if-else Statement

```
In [14]: if True:
    print('Data Science')
else:
    print('Bye for now')
```

Data Science

```
In [15]: if False:
    print('Data Science job')
else:
    print('No job')
```

No job

```
In [16]: x = 8
r = x % 2

if r == 0:
    print('Even number')

else:
    print('Odd number')
```

Even number

```
In [17]: x = 9
r = x % 2
```

```
if r != 0:  
    print('Odd number')  
  
else:  
    print('Even number')
```

Odd number

```
In [18]: x = 3  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
             if x > 5:  
                 print('Greater number')  
         else:  
             print('Odd number')
```

Odd number

```
In [19]: x = 4  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
             if x > 5:  
                 print('Greater number')  
         else:  
             print('Odd number')
```

Even number

```
In [20]: x = 4  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
  
             if x > 5:  
                 print('Greater number')  
             else:  
                 print('Lesser number')  
         else:  
             print('Odd number')
```

Even number

Lesser number

```
In [21]: x = 13  
         r = x % 2  
  
         if r == 0:  
             print('Even number')  
  
             if x > 5:  
                 print('Greater number')  
             else:  
                 print('Lesser number')
```

```
else:  
    print('Odd number')
```

Odd number

```
In [22]: x = 3  
  
if x ==1:  
    print('One')  
if x == 2:  
    print('Two')  
if x == 3:  
    print('Three')  
if x == 4:  
    print('four')
```

Three

(iii) if elif else Statement

```
In [23]: x = 1  
  
if x ==1:  
    print('One')  
elif x == 2:  
    print('Two')  
elif x == 3:  
    print('Three')  
elif x == 4:  
    print('four')
```

One

```
In [24]: x = 5  
  
if x ==1:  
    print('One')  
  
elif x == 2:  
    print('Two')  
elif x == 3:  
    print('Three')  
elif x == 4:  
    print('four')  
  
else:  
    print('Number is not found')
```

Number is not found

```
In [25]: age = 19  
if age > 18:  
    print("Eligible to Vote")
```

Eligible to Vote

Loops

```
In [26]: print('Data Science')
print('Data Science')
print('Data Science')
print('Data Science')
print('Data Science')
```

Data Science
Data Science
Data Science
Data Science
Data Science

(i) while loop

```
In [27]: i = 1 #Initializing

while i <= 5: #Condition
    print('Data Science')
    i = i + 1 #Increment
```

Data Science
Data Science
Data Science
Data Science
Data Science

```
In [28]: i = 5 #Initializing
while i >= 1: #Condition
    print('Data Science')
    i = i - 1 #Decrement
```

Data Science
Data Science
Data Science
Data Science
Data Science

```
In [29]: i = 5 #Initializing
while i >= 1: #Condition
    print('Data Science',i)
    i = i - 1 #Decrement
```

Data Science 5
Data Science 4
Data Science 3
Data Science 2
Data Science 1

```
In [30]: i = 1 #Initializing
while i <= 5: #Condition
    print('Data Science',i)
    i = i + 1 #Increment
```

Data Science 1
Data Science 2
Data Science 3
Data Science 4
Data Science 5

```
In [31]: i = 1
while i <= 5:
    print('Data Science') #When we mention end then new line will not create

    j = 1
    while j <= 4:
        print('Technology')
        j = j + 1

    i = i + 1
    print()

#The output which we got is very lengthy but how to make them on eline Lets
```

Data Science
Technology
Technology
Technology
Technology

Data Science
Technology
Technology
Technology
Technology

Data Science
Technology
Technology
Technology
Technology

Data Science
Technology
Technology
Technology
Technology

Data Science
Technology
Technology
Technology
Technology

```
In [32]: i = 1
while i <= 5:
    print('Data Science', end = ' - ') #When we mention end then new line will

    j = 1
    while j <= 4:
        print('Technology', end = ' - ')
        j = j + 1

    i = i + 1
    print()

#The output which we got isn't Lengthy as the previous one
```

```
Data Science - Technology - Technology - Technology - Technology -
Data Science - Technology - Technology - Technology - Technology -
Data Science - Technology - Technology - Technology - Technology -
Data Science - Technology - Technology - Technology - Technology -
Data Science - Technology - Technology - Technology - Technology -
```

```
In [36]: i = 1

while i <= 5:
    print('Data Science', end = " ") #When we mention end then new line will no

    j = 1
    while j <= 4:
        print('Technology', end = " ")
        j = j + 1

    i = i + 1
    print()

#The output which we got isn't lengthy as the previous one
```

```
Data Science Technology Technology Technology Technology
Data Science Technology Technology Technology Technology
Data Science Technology Technology Technology Technology
Data Science Technology Technology Technology Technology
Data Science Technology Technology Technology Technology
```

```
In [37]: # Lets use 'while Loop' using some numbers
i = 1

while i <= 2:
    j = 0
    while j <= 2:
        print(i*j, end = " ")
        j += 1
    print()
    i +=1
```

```
0 1 2
0 2 4
```

```
In [38]: # Lets use 'while Loop' using some numbers
i = 1
while i <= 4:
    j = 0
    while j <= 3:
        print(i*j, end = " ")
        j +=1
    print()
    i += 1
```

```
0 1 2 3
0 2 4 6
0 3 6 9
0 4 8 12
```

(ii) for loop

```
In [1]: name = 'nit'
```

```
for i in name:  
    print(name)
```

nit
nit
nit

In [2]: name = 'nit'

```
for i in name:  
    print(i)
```

n
i
t

In [3]: l = [1,2,3, 'nit']

In [4]: name1 = [1, 3.5, 'Hello'] *# I want to to print the value individually*

```
for i in name1:  
    print(i)
```

1
3.5
Hello

In [5]: for i in [2, 3, 7.8, 'Hi']:
 print(i)

2
3
7.8
Hi

In [6]: range(5)

Out[6]: range(0, 5)

In [7]: for i in range(5):
 print(i)

0
1
2
3
4

In [8]: for i in range(2,5):
 print(i)

2
3
4

In [10]: for i in range(1, 10, 3):
 print(i)

1
4
7

```
In [9]: # Print the value which is divisible by 5 table  
for i in range(1,21):  
    print(i)
```

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

```
In [11]: # Print the value which is divisble by 5
```

```
for i in range(1,51):  
    if i % 5 == 0:  
        print(i)
```

5
10
15
20
25
30
35
40
45
50

```
In [12]: # Print the value which is divisible by 5, i don't want that value
```

```
for i in range(1,51):  
    if i % 5 != 0:  
        print(i)
```

1
2
3
4
6
7
8
9
11
12
13
14
16
17
18
19
21
22
23
24
26
27
28
29
31
32
33
34
36
37
38
39
41
42
43
44
46
47
48
49

In [39]: *# Write the code user ask chocolate from vendor, machine write the basic code*

```
x = int(input('How many chocolates you want ?'))  
  
i = 1  
while i <= x:  
    print('Chocolate')  
    i += 1
```

Chocolate
Chocolate
Chocolate
Chocolate

In [40]: *ava = 5 # The machine has only 5 chocolates*

```
x = int(input('How many chocolates you want ?'))  
  
i = 1
```

```
while i <= x:
    print('Chocolate')
    i += 1
```

*# If you check the user wants 10 chocolates but available chocolate is 5 but we
In this code we just declare but we don't apply any condition to it*

Chocolate
Chocolate
Chocolate
Chocolate
Chocolate
Chocolate

```
In [43]: available_chocolate = 5 # The machine has only 5 chocolates

x = int(input('How many chocolates you want ?'))

i = 1
while i <= x:
    if i > available_chocolate: # We stop the execution but which code executi
        break # break is statement | means jump out of the loop
    print('Chocolate')
    i += 1

print('Bye for now')
```

Chocolate
Chocolate
Chocolate
Chocolate
Bye for now

```
In [44]: available_chocolate = 5 # The machine has only 5 chocolates

x = int(input('How many chocolates you want ?'))

i = 1
while i <= x:

    if i > available_chocolate: # We stop the execution but which code executi
        print('Out of stock')
        break # break is statement | means jump out of the loop
    print('Chocolate')
    i += 1

print('Bye for now')
```

Chocolate
Chocolate
Chocolate
Chocolate
Chocolate
Out of stock
Bye for now

```
In [13]: for i in range(1,11):
          print(i)
```

```
1
2
3
4
5
6
7
8
9
10
```

```
In [56]: # BREAK Statement - It is used when you want to break the Loop
         # I don't want to print 11 no., i just want only 5 no. of range from 1 to 10
```

```
In [14]: for i in range(1,11):
         if i == 5:
             break
         print(i)
```

```
1
2
3
4
```

```
In [ ]: #
```

```
In [16]: for i in range(1,11):
         if i == 10:
             break
         print(i)
```

```
1
2
3
4
5
6
7
8
9
```

```
In [15]: for i in range(1,51):
         if i % 3 == 0:
             print(i)
         print('end')
```

```
3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48
end
```

In [53]: *# CONTINUE Statement - It won't be terminate & exclude the assign number, it giv*

```
In [15]: for i in range(1,51):
          if i % 3 == 0:
              continue
          print(i)
          print('end')
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50
end
```

```
In [16]: for i in range(1,51):
          if i % 3 == 0 or i % 5 == 0:
              continue
          print(i)
          print('end')
```

```
1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
end
```

```
In [1]: for i in range(1,11):
        if i == 8:
            continue
        print(i)
```

```
1
2
3
4
5
6
7
9
10
```

```
In [2]: for i in range(1,11):
        if i == 6:
            continue
        print('Hello :',i)
```

```
Hello : 1
Hello : 2
Hello : 3
Hello : 4
Hello : 5
Hello : 7
Hello : 8
Hello : 9
Hello : 10
```

```
In [1]: for i in range(1, 11):
```

```
Cell In[1], line 1
    for i in range(1, 11):
        ^
_IncompleteInputError: incomplete input
```

```
In [51]: # PASS Statement - Pass the code & it won't go (Code give you the error)
```

```
In [2]: for i in range(1,11):
        pass
```

How to print the numbers which is not divisible by 3 or 5 between 1 to 50 ?

```
In [3]: for i in range(1,51):
        if i % 3 == 0:
            print(i)
        print('end')
```

```
3
6
9
12
15
18
21
24
27
30
33
36
39
42
45
48
end
```

```
In [4]: for i in range(1,51):
        if i % 3 == 0:
            continue
        print(i)
        print('end')
```

```
1
2
4
5
7
8
10
11
13
14
16
17
19
20
22
23
25
26
28
29
31
32
34
35
37
38
40
41
43
44
46
47
49
50
end
```

```
In [50]: for i in range(1,51):
          if i % 3 == 0 or i % 5 == 0:
              continue
          print(i)
          print('end')

          #print('end')
          # It will skip all the value which is divisible by 3 or 5
```

```
1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
end
```

```
In [6]: # When you apply, you won't get the value which is divisible by both 3 & 5 (15)

for i in range(1,50):
    if i % 3 == 0 or i % 5 == 0:
        continue
    print(i)
print('end')
```

```
1
2
4
7
8
11
13
14
16
17
19
22
23
26
28
29
31
32
34
37
38
41
43
44
46
47
49
end
```

In [7]: *# I don't want to print the values which are even numbers that means print only*

```
for i in range(1,51):
    if (i % 2 == 0):
        #print('Even')
        continue
    else:
        print(i)
print('Bye!!!')
```

1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
39
41
43
45
47
49
Bye!!

Printing Pattern in Python

```
In [9]: print('# # # #')
        print('# # # #')
        print('# # # #')
        print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [17]: for i in range(1,5):
        i = i+1
        print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [19]: for i in range(1,5):
        if i <= 5:
            print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [10]: for j in range(4):
        print('#')
```

```
#
#
#
#
```

```
In [11]: for j in range(4):
          print('# # # #')
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [12]: for j in range(4):
          print('#', end = " ")
```

```
# # # #
```

```
In [13]: for j in range(4):
          print('#', end = " ")

          for j in range(4):
              print('#', end = " ")
```

```
# # # # # # # #
```

```
In [14]: for j in range(4):
          print('#', end = " ")

          print()

          for j in range(4):
              print('#', end = " ")
```

```
# # # #
# # # #
```

```
In [49]: for j in range(4):
          print('#', end = " ")

          print()

          for j in range(4):
              print('#', end = " ")

          print()

          for j in range(4):
              print('#', end = " ")

          print()

          for j in range(4):
              print('#', end = " ")
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [ ]: # We should use Debug mode in Pycharm
```

```
In [17]: for i in range(4):
          for j in range(4):
              print('#', end = " ")
          print()
```

```
# # # #
# # # #
# # # #
# # # #
```

```
In [22]: for i in range(4):
          for j in range(i+1):
              print('#', end = ' ')
          print()
```

```
#
# #
# # #
# # # #
```

```
In [19]: for i in range(1,5):
          print('# '*i)
```

```
#
# #
# # #
# # # #
```

```
In [20]: for i in range(1,5):
          for j in range(4):
              if i>j:
                  print("#", end = " ")
          print()
```

```
#
# #
# # #
# # # #
```

```
In [21]: list(range(5))
```

```
Out[21]: [0, 1, 2, 3, 4]
```

```
In [22]: for i in range(4):
          for j in range(i):
              print('#', end = " ")
          print()
```

```
#
# #
# # #
```

```
In [23]: for i in range(4):
          for j in range(i+1):
              print('#', end = " ")
          print()
```

```
#
# #
# # #
# # # #
```

In []:

```
In [48]: for i in range(4):
        for j in range(4-i):
            print('#', end = " ")
        print()
```

```
# # # #
# # #
# #
#
```

```
In [25]: for i in range(1,5):
        print("# "*(5-i))
```

```
# # # #
# # #
# #
#
```

```
In [27]: nums = [7, 14, 18, 21, 23, 27]
        for num in nums:
            if num % 5 == 0:
                print(num)
                break
        else:
            print('Number is not found')
```

Number is not found

(iii) for else loop

```
In [26]: nums = [12,15,18,21,26]

        for num in nums:
            if num % 5 == 0:
                print(num)
```

15

```
In [27]: nums = [12,14,18,21,25,30,35]

        for num in nums:
            if num % 5 == 0:
                print(num)
```

25

30

35

```
In [28]: nums = [12,14,18,21,25,20]

        for num in nums:
            if num % 5 == 0:
                print(num)
```

25

20

```
In [29]: nums = [12,14,18,21,20,25]

for num in nums:
    if num % 5 == 0:
        print(num)
        break
```

20

```
In [30]: nums = [10,14,18,21,5,10]

for num in nums:
    if num % 5 == 0:
        print(num)
        break      # It'll print only 1 number then it break
```

10

```
In [31]: nums = [7,14,18,21,23,27]      # Here, there is no number which is divisible by 5

for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Number Not Found')      # Here, else we don't write in if block but
```

Number Not Found

```
In [40]: nums = [7,14,18,21,23,27,29]      # Here, there is no number which is divisible by

for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Number Not Found')      # Every iteration it checking condition
```

Number Not Found
 Number Not Found
 Number Not Found
 Number Not Found
 Number Not Found
 Number Not Found
 Number Not Found

```
In [47]: nums = [7,14]

for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Number Not Found')      # Every iteration it checking condition
```

Number Not Found
 Number Not Found

```
In [43]: nums = [7,14,18,21,23,27]      # Here, there is no number which is divisible by 5

for num in nums:
    if num % 5 == 0:
```

```

        print(num)
        break
    else:
        print('Number Not Found')    # Here, else we don't write in if block but

```

Number Not Found

```

In [32]: nums = [10,14,18,21,20,27]    # Here, there is no number which is divisible by 5

for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Not Found')

```

10

```

In [44]: nums = [60,14,18,21,20,27]    # Here, there is no number which is divisible by 5

for num in nums:
    if num % 5 == 0:
        print(num)
        #break
    else:
        print('Not Found')

```

60

20

Not Found

```

In [46]: nums = [70,14,18,21,20,27]    # Here, there is no number which is divisible by 5

for num in nums:
    if num % 5 == 0:
        print(num)
        break
    else:
        print('Not Found')

```

70

How to check given number is prime number or not ?

```

In [34]: num = 13

for i in range(2,num):
    if num % i == 0:
        print('Not Prime Number')
        break
    else:
        print('Prime Number')

```

Prime Number

```

In [35]: num = 25

for i in range(2, num):
    if num % i == 0:
        print('Not Prime Number')
        break

```

```
else:  
    print('Prime Number')
```

Not Prime Number